

Benchmarking Neural Surrogates for Hydrofoil RANS Flow and Hydrodynamic Shape Optimization

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Abstract—Neural flow surrogates are usually compared by field error, although their practical value often lies in the designs they select. We compare a convolutional neural network (CNN) U-Net, a physics-regularized point network, a Fourier neural operator (FNO), and a deep operator network (DeepONet) on 381 OpenFOAM simulations of hydrofoils in water. Complete NACA families form disjoint training, validation, and final-test sets, and every architecture is trained with three seeds. The models predict Reynolds-averaged Navier–Stokes (RANS) fields, lift, drag, and pressure-based cavitation-inception risk before driving the same four-parameter shape search. DeepONet gives the lowest mean pressure, C_p , lift, and drag errors on the untouched test families; the physics-regularized network is close while using one tenth as many parameters. Inference is 384–867 times faster than CFD, replacing repeated inner-loop solves while retaining CFD endpoint checks. Across seeds, fresh OpenFOAM endpoint solves give mean lift-to-drag ratios from 22.10 to 37.90, but a three-level grid study shows that drag remains mesh-sensitive. Field accuracy, force accuracy, search quality, and numerical verification are therefore related but not interchangeable.

Index Terms—artificial intelligence for science, CFD surrogate, neural operator, physics-informed learning, hydrofoil optimization, cavitation inception

I. Introduction

For a hydrofoil operating in water, shape design is an optimization problem wrapped around a flow solver. Each change in camber, thickness, or incidence requires another Reynolds-averaged Navier–Stokes (RANS) solution, so even a modest search can consume hundreds of computational fluid dynamics (CFD) evaluations; neural surrogates move most of that search off the solver by learning the map from geometry and operating conditions to flow and forces [1]. The difficult question is not simply which network minimizes field error, but which one preserves the quantities that determine a useful design.

The convolutional neural network U-Net (CNN-U-Net), Fourier neural operator (FNO), deep operator network (DeepONet), and physics-regularized point network impose very different structure on that map [2], [3], [4], [5]. A convolutional model emphasizes local multiscale features; FNO mixes information globally in Fourier space; DeepONet separates parameter and coordinate representations; and a physics penalty constrains a pointwise regressor. Their relative merits may therefore change as the target moves from pressure fields to forces and finally to shape optimization.

Airfoil-surrogate research has progressed from field prediction [6], [7], [8], [9] to benchmarks that evaluate surface forces, out-of-distribution accuracy, and computational cost [10], [11], [12]. Those metrics establish whether a model approximates CFD, but they do not by themselves establish whether it will make the same design decision. A small, structured force error can be amplified when an optimizer repeatedly queries a narrow region of design space. The missing experiment is a controlled comparison that follows several surrogate families through the same optimizer and then checks the selected geometries with new CFD solutions. High-fidelity endpoint verification is standard in surrogate shape optimization, while a matched direct-CFD search provides a stronger reference when its cost is manageable [13], [14].

This setting is relevant to high-performance simulation workflows even though the present benchmark runs locally: independent CFD cases can be generated in parallel, the resulting ensemble trains a reusable surrogate, and batched inference replaces most solver calls in sweeps or design loops while selected designs return to CFD for verification.

That experiment is the contribution of this paper; we do not propose a new architecture. We compare four representative families on the same RANS data, complete-geometry validation split, targets, and three-start hydrodynamic shape optimization (HSO) problem. The benchmark connects field and force accuracy, pressure-threshold cavitation screening, and runtime to the final geometry selected by each model. Fresh OpenFOAM solutions measure endpoint discrepancy, and a direct OpenFOAM search uses the same starts and optimization budget. Together, these tests ask whether validation accuracy predicts search quality and how closely each inexpensive search approaches the CFD-driven result.

II. Related Work

Convolutional surrogates established that steady flow fields can be learned directly from geometry encodings on Cartesian grids [6], [7], [9]. Later datasets and benchmarks broadened evaluation beyond image error: AIRFRANS includes surface quantities and out-of-distribution splits, the geometric neural operator (GINO) addresses irregular three-dimensional geometries, and ML4CFD evaluates

modern operator-learning systems under a common protocol [10], [11], [12]. These studies motivate our geometry-level validation split and separate reporting of fields, forces, runtime, and low-pressure classification.

Neural operators represent an entire parameter-to-solution map rather than a single flow realization. FNO performs global spectral mixing on a regular grid, whereas DeepONet combines a parameter-dependent branch representation with coordinate-dependent trunk functions [3], [4], [15]. Physics-informed learning offers a different inductive bias by penalizing differential-equation residuals [5], [16]. The present comparison intentionally includes both grid operators and pointwise models, since their relative value may change when the output is used for optimization rather than visualization.

Learning-based airfoil optimization has shown large online speedups and has validated selected optima with higher-fidelity solvers [13], [14]. Most such studies develop or test one surrogate family. Our contribution is complementary: data, split, targets, optimizer, starts, bounds, and endpoint solver are held fixed while the model family changes. A matched direct-CFD search then distinguishes prediction error from the effect of optimizing a learned objective.

III. Data and CFD Reference

A. Design space

The dataset contains National Advisory Committee for Aeronautics (NACA) 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418, and 4421 profiles generated from the standard four-digit parameterization [17]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$ and Reynolds numbers $Re \in \{10^5, 2 \times 10^5, 5 \times 10^5, 10^6\}$. All cases use liquid water with density 997 kg/m^3 and kinematic viscosity $10^{-6} \text{ m}^2/\text{s}$. With a 1 m chord, the freestream speed is set by $U_\infty = Rev/c$; the far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

Each case is solved with OpenFOAM simpleFoam [18]. For steady incompressible flow, the mean velocity $\bar{u}_i$ and pressure $\bar{p}$ satisfy

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (1)$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \rho \frac{\partial \overline{u'_i u'_j}}{\partial x_j}. \quad (2)$$

The Reynolds stress is closed with the k - ω shear-stress transport (SST) model [19]. Angled freestream velocity is imposed at the inlet and far boundary, and OpenFOAM's forceCoeffs supplies total lift and drag after 300 iterations. Cell-centered fields are interpolated to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs include coordinates, operating parameters, three NACA descriptors, a fluid mask, and signed distance to the profile.

Three simulations with $|C_L|$ or $|C_D| > 5$ are discarded before splitting, leaving 381 cases. The split is fixed before training and is identical for all architectures and seeds. NACA 0015 and 2412 provide 64 validation cases for early stopping; NACA 0018 and 4418 provide an untouched 64-case test set used only for final reporting. The remaining 253 cases from eight families train the models. No validation or test geometry therefore appears in training at a different operating condition. All retained solves reach iteration 300. At the final iteration, the 95th-percentile maximum initial equation residual is 1.27×10^{-4} ; over the last 50 iterations, the 95th-percentile spans are 0.0059 for C_L and 2.65×10^{-4} for C_D .

B. Targets and cavitation-risk definition

The common output vector contains U_x , U_y , absolute pressure p , eddy viscosity, k , ω , pressure coefficient C_p , cavitation margin, lift coefficient C_L , and drag coefficient C_D . Grid fields use masked normalized mean-squared error, while the case-level force targets use normalized mean-squared error. OpenFOAM's kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \quad C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2}. \quad (3)$$

For an ambient-pressure value p_a , the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y) \frac{1}{2} \rho U_\infty^2 - p_v. \quad (4)$$

A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in \{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test from the predicted C_p field without representing multiphase physics.

IV. Surrogates and Optimization

A. Model families

The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization, Gaussian error linear unit (GELU) activations, and base width 16. It maps the complete input grid to the output grid [2]. The FNO has width 32, four operator blocks, and 12 retained Fourier modes in each direction. One block is

$$h_{\ell+1} = \sigma(W_\ell h_\ell + \mathcal{F}^{-1}(R_\ell \mathcal{F}(h_\ell))). \quad (5)$$

following the spectral operator construction of [3]. Here, W_ℓ is a local linear map and R_ℓ acts on the retained spectral modes. The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions. For output k ,

$$\hat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \frac{\mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x})}{\sqrt{64}} + c_k, \quad (6)$$

as in the branch-trunk construction of [4], where $\boldsymbol{\mu}$ contains the operating and geometry parameters.

The physics-regularized multilayer perceptron (MLP) has three tanh layers of width 64. It is supervised by

TABLE I

Surrogate representations and measured inference cost per case.

Model	Representation	Principal bias	Params.	ms
CNN-U-Net	grid	local, multiscale	485,178	10.80
Physics-reg. MLP	points	differential residual	9,930	6.88
FNO	grid	global, spectral	595,914	12.07
DeepONet	branch-trunk	operator basis	100,746	13.52

CFD values at 1,024 sampled points per case. Its extra loss uses the directional residuals

$$r_c = u_x + v_y, \quad r_x = uu_x + p_x/\rho, \quad r_y = vv_y + p_y/\rho, \quad (7)$$

with $\mathcal{L} = \mathcal{L}_{\text{sup}} + 10^{-4}\mathcal{L}_{\text{phys}}$, following the residual-loss principle in [5]. This is a lightweight physics regularizer, not a data-free physics-informed neural network (PINN) or a complete RANS residual.

Table I summarizes the resulting inductive biases. The two grid models consume the entire Cartesian field at once. DeepONet and the physics-regularized MLP instead evaluate coordinates, which makes their representation naturally pointwise, although the present timing benchmark requests the same complete grid from every model.

All models use AdamW with learning rate 2×10^{-3} , weight decay 10^{-4} , at most 120 epochs, patience 20, and gradient clipping. We train each architecture with seeds 7, 17, and 27, always using the same family split. Normalization is computed from training cases only. The best checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters (Table II).

The comparison standardizes data, targets, optimizer, stopping rule, and evaluation budget, but does not force equal parameter counts. Equalizing width would not equalize effective capacity across convolutional, spectral, branch-trunk, and pointwise representations. We therefore report parameter count, training time, and inference time explicitly and interpret the study as a comparison of compact canonical configurations, not a capacity-matched architecture ablation. No hyperparameter is selected on the test families.

Figure 1 reports the complete optimization histories rather than only the selected checkpoints. Because the architectures consume the data in different forms and the objective aggregates heterogeneous normalized targets, loss magnitude should be interpreted within a model, not as a direct cross-model accuracy metric. DeepONet reaches its minimum validation objective at epoch 6, the physics-regularized network at epoch 30, FNO at epoch 54, and CNN-U-Net at epoch 120. The isolated FNO spikes are retained in the record and motivate reporting best-checkpoint results alongside the full history.

Optimization uses dedicated C_L and C_D output heads supervised by OpenFOAM’s total force coefficients. Pressure integration on the coarse common grid is not used for the objective because it omits viscous traction and under-resolves the wall.

B. Why physics regularization rather than a full PINN?

A classical PINN is trained by differentiating its outputs and minimizing the governing-equation and boundary-condition residuals, sometimes with few or no solution labels [5]. Our point network remains primarily supervised by OpenFOAM values; its low-weight residual contains continuity and inviscid directional momentum only. It omits viscous stress, Reynolds stress, the k - ω transport equations, and explicit wall and far-field residuals. Describing it as a full RANS PINN would therefore overstate the physics imposed during training.

This lighter comparator was chosen because it applies a recognizable physical bias while retaining the same labels, split, and modest training budget as the other surrogates. A full hydrofoil RANS PINN is a different experiment: it would require coupled turbulence variables and boundary losses, careful residual balancing near the wall, and substantially more automatic-differentiation work. The label “physics-regularized MLP” keeps that distinction visible when interpreting its strong pressure results.

C. Common optimization protocol

At $Re = 5 \times 10^5$, each surrogate drives the same HSO loop. The design vector $\theta = (m, p, t, \alpha)$ contains maximum camber, camber location, thickness, and incidence, with bounds $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and $\alpha \in [-4, 10]^\circ$. We launch Nelder–Mead [20] from NACA 0012, 2412, and 4415 and give every run the same 40-iteration budget. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}}. \quad (8)$$

The search calls only the surrogate; OpenFOAM is not used inside the loop. Cavitation margin is recorded afterward as a secondary diagnostic and does not alter the objective. We return the best design observed within the fixed budget, select the strongest of the three starts for each architecture, and then generate a new mesh and independent OpenFOAM simulation for that winner.

Although training geometries are discrete NACA families, the model inputs contain continuous four-digit descriptors (m, p, t) , and the optimizer queries intermediate values within the stated bounds. Some queries extrapolate beyond the convex coverage of the training shapes, especially near a thickness bound. We therefore retain every evaluation in the trace and independently remesh and solve each selected endpoint with CFD. This validates the reported design, not the full learned objective surface.

For the direct-CFD baseline, the same Nelder–Mead procedure starts from the same three foils and uses the same bounds, objective, tolerances, and 40-iteration budget, but every evaluation is obtained by remeshing and running OpenFOAM. This separates endpoint verification, which checks each surrogate’s winner once, from CFD-

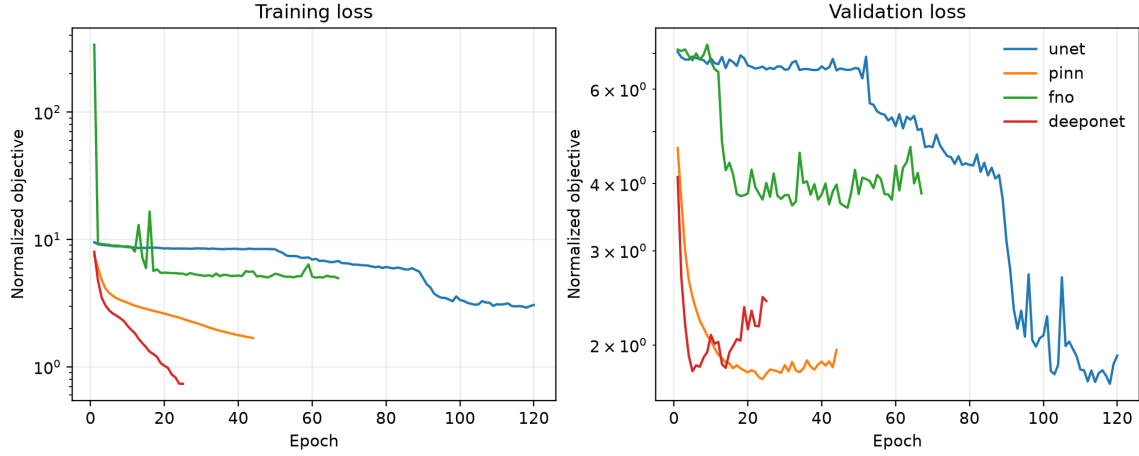


Fig. 1. Training and unseen-geometry validation objectives. Curves terminate after early stopping; objectives combine normalized field and force losses.

TABLE II
Final-test RMSE (mean $\pm$ sample standard deviation over three seeds), size, and speedup.

Model	p (Pa)	C_p	C_L	C_D	Params.	Speedup
CNN-U-Net	28.32 ± 14.18	$0.0657 \pm .0373$	$0.258 \pm .200$	$0.00871 \pm .00522$	485,178	544 ± 13
Physics-reg. MLP	11.56 ± 0.48	$0.0445 \pm .0012$	$0.0352 \pm .0056$	$0.00328 \pm .00029$	9,930	867 ± 1
FNO	27.79 ± 1.07	$0.0643 \pm .0040$	$0.248 \pm .020$	$0.0103 \pm .00054$	595,914	464 ± 6
DeepONet	10.74 ± 1.48	$0.0411 \pm .0045$	$0.0304 \pm .0081$	$0.00339 \pm .00089$	100,746	384 ± 2

driven optimization, which supplies a matched reference search.

V. Results

A. Validation accuracy and computational cost

Table II reports mean and one sample standard deviation over three independently trained models on the untouched 64-case NACA 0018/4418 test set. DeepONet gives the lowest mean errors for pressure, C_p , C_L , and C_D ; the physics-regularized network is consistently second and varies less across seeds. CNN-U-Net is seed-sensitive, especially for the global force heads, while FNO retains weaker force trends in this low-geometry regime.

Mean training times are 556, 67, 270, and 47 s for CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively. Median OpenFOAM execution time is 12.4 s per case, while complete-field inference requires 14–32 ms, yielding 384–867 $\times$ speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch 2.13 ran on its central processing unit (CPU), and single-process OpenFOAM 9 ran in Docker. These numbers measure repeated local inference and exclude data generation and mesh construction.

The ranking also depends strongly on the target field. Figure 3 shows that the pointwise models recover both velocity components and pressure, whereas CNN-U-Net and FNO are substantially weaker on absolute pressure and U_x . All four models have $R^2 \leq 0.15$ for eddy viscosity, k , and ω . Thus, good C_p and force predictions do not make

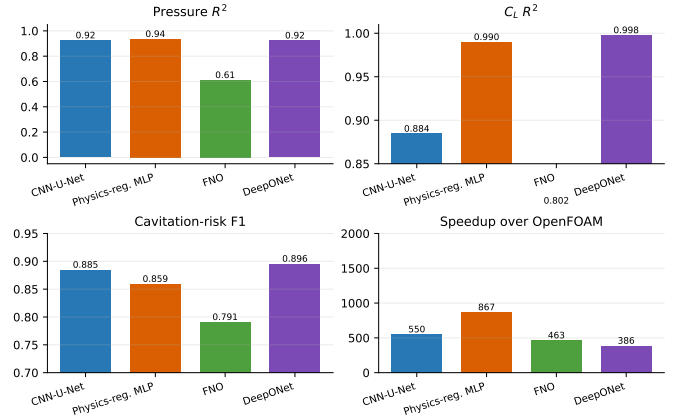


Fig. 2. No architecture dominates every criterion on the 64-case final test set (seed 7 shown for field-level detail).

these checkpoints interchangeable with a complete RANS solution; their demonstrated value is in pressure screening and force-guided design.

Figure 4 shows a representative final-test case from the unseen NACA 0018 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. A common color scale makes the location and magnitude of those differences directly comparable across all five panels.

B. Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so we report F1 rather than accuracy. Aggregated over the ambient-pressure sweep, DeepONet obtains 0.900 ± 0.008 , the physics-regularized network 0.867 ± 0.008 , CNN-U-Net 0.758 ± 0.198 , and FNO 0.724 ± 0.065 . The large CNN-U-Net spread mirrors its field and force variability. This secondary diagnostic asks whether a surrogate preserves

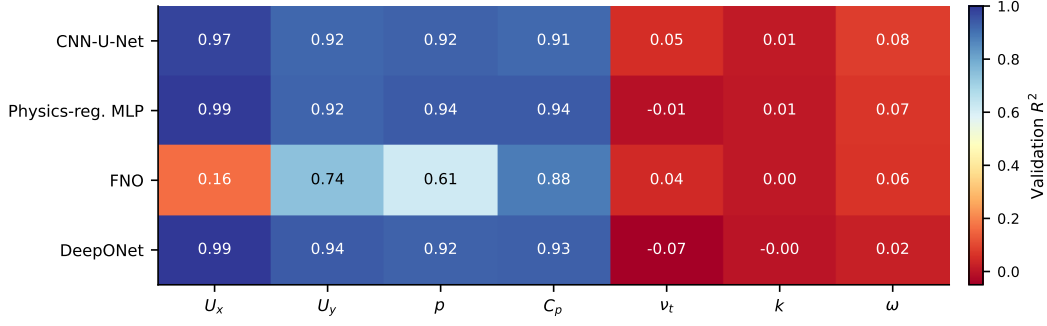


Fig. 3. Full-field coefficient of determination on the unseen-family test set (seed 7). The turbulence outputs remain difficult for every model even when velocity, pressure, or C_p is accurate.

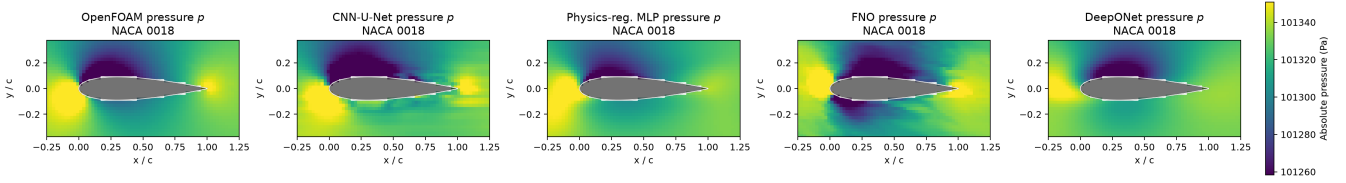


Fig. 4. Absolute-pressure prediction for a representative NACA 0018 test case (case 179), shown over $x/c \in [-0.25, 1.25]$ and $y/c \in [-0.375, 0.375]$. Every panel uses the same color scale.

TABLE III
Pressure-threshold diagnostic over the final test set (mean $\pm$ sample standard deviation over three seeds).

Model	Precision	Recall	F1
CNN-U-Net	$0.817 \pm .084$	$0.730 \pm .274$	$0.758 \pm .198$
Physics-reg. MLP	$0.959 \pm .018$	$0.791 \pm .023$	$0.867 \pm .008$
FNO	$0.615 \pm .106$	$0.892 \pm .024$	$0.724 \pm .065$
DeepONet	$0.955 \pm .001$	$0.851 \pm .014$	$0.900 \pm .008$

low- C_p regions; it does not model vapor fraction, cavity length, or shedding [21].

C. Shape optimization against direct CFD

Table V reports the complete L/D comparison: the OpenFOAM value of the starting foil, the surrogate estimate at its selected design, and the result of remeshing that design and solving it once in OpenFOAM. That one-solve check is endpoint validation. Separately, the direct baseline places OpenFOAM inside the matched three-start optimizer, providing a CFD-driven reference under the same local-search budget.

Optimization is also seed-dependent. Across the three selected endpoints per architecture, medium-grid OpenFOAM gives $L/D = 24.44 \pm 1.35$ for CNN-U-Net, 37.90 ± 1.59 for the physics-regularized network, 22.10 ± 3.46 for FNO, and 37.47 ± 5.95 for DeepONet. The matched direct-CFD search reaches 17.80, 27.33, and 29.89 from NACA 0012, 2412, and 4415, respectively. These endpoint means describe the old medium discretization; the grid study

below is needed before treating their differences as mesh-independent.

VI. Discussion

The repeated runs change the interpretation of the benchmark. DeepONet is the best final-test predictor, yet its selected L/D varies by 5.95 across seeds; the physics-regularized network has a slightly lower mean endpoint but only 1.59 standard deviation. CNN-U-Net is unstable in both force prediction and optimization, while FNO is more repeatable but settles on lower-performing designs. Predictive accuracy, optimizer ranking, and seed robustness are thus three separate criteria.

Figure 6 shows why all starts matter. The two cambered starts often enter different basins from NACA 0012, and DeepONet’s learned objective is particularly sharp. The direct run remains a matched local-search baseline, not a global CFD optimum. Surrogate and direct traces should therefore be read as budgeted search behavior rather than proof that one evaluator dominates the continuous design space.

The grid check is the most important qualification. All 12 selected-design solves converge iteratively by 1,000 iterations, with final residuals below 10^{-7} and negligible late-window force variation. Lift changes by at most about 4% from the medium to fine mesh, but drag changes by 16–24%, producing 16–20% shifts in L/D . Consequently, the endpoint ordering is evidence about this CFD workflow, not a mesh-independent hydrodynamic optimum.

TABLE IV

Best L/D objective found from each starting foil. Surrogate rows report learned objective values inside their search; direct CFD reports OpenFOAM values. Bold entries are the starts selected for endpoint validation.

Evaluator	NACA 0012 start	NACA 2412 start	NACA 4415 start
CNN-U-Net	23.47	22.65	25.82
Physics-reg. MLP	21.77	33.27	32.68
FNO	18.57	18.77	20.80
DeepONet	19.91	60.94	59.82
Direct CFD	17.80	27.33	29.89

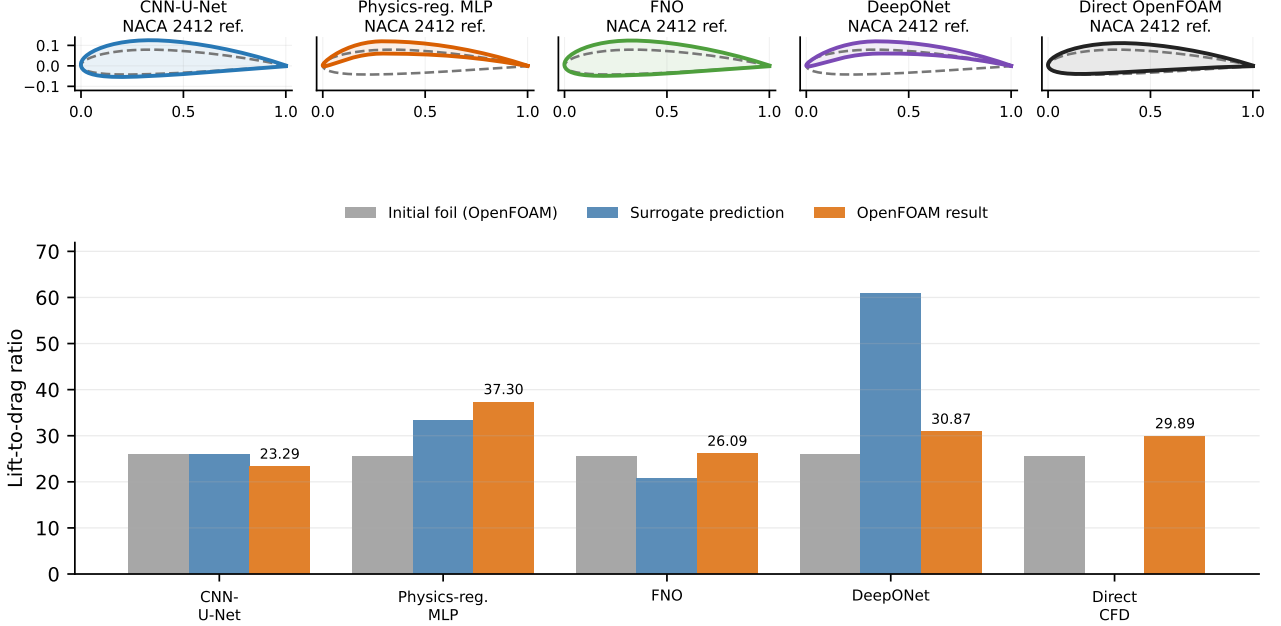


Fig. 5. Top: selected shapes against the same dashed NACA 2412 geometric reference. The common reference supports shape comparison and is not necessarily the optimizer initializer; actual initializers are listed in Table IV. Bottom: initial-foil OpenFOAM values, surrogate predictions at the selected designs, fresh OpenFOAM endpoint checks, and the direct-CFD result.

TABLE V

HSO endpoints over three training seeds. OpenFOAM columns report the medium-grid mean and sample standard deviation. Grid shift is the seed-7 medium-to-fine L/D change relative to the fine result.

Model	Pred. mean	CFD mean	CFD std.	Grid shift
CNN-U-Net	21.75	24.44	1.35	18.9%
Physics-reg. MLP	38.82	37.90	1.59	19.6%
FNO	20.93	22.10	3.46	15.7%
DeepONet	51.08	37.47	5.95	19.4%
Direct CFD	—	29.89	—	—

A boundary-layer mesh and a formal asymptotic grid-convergence study are required before design deployment.

The field results help explain why parameter count alone is not predictive. The 9,930-parameter physics-regularized MLP is the smallest model and nearly matches the best pressure result, while the 595,914-parameter FNO is largest and ranks third on pressure. In this dataset, an operator model receives only 253 complete training realizations even though each realization contains

many grid cells. The pointwise networks turn those same cases into supervised coordinate samples and encode the operating parameters directly. This is favorable for a fixed grid and a low number of geometries, but it does not imply that pointwise models will dominate as geometric diversity, resolution, or three-dimensional domain size grows. The comparison instead identifies the inductive bias that works best at the present data scale.

Runtime should likewise be interpreted at the workflow level. The 384–867 fold online speedups exclude CFD data generation, which is the dominant up-front expense and must be amortized across operating sweeps or design studies. Case generation is embarrassingly parallel, and inference can be batched across candidate geometries, making the pipeline compatible with high-performance parameter-sweep and optimization workflows. We did not benchmark distributed training, multi-node CFD, or parallel scaling, so these results do not constitute a claim of cluster-level acceleration or replace the simulations needed to construct and verify the surrogate.

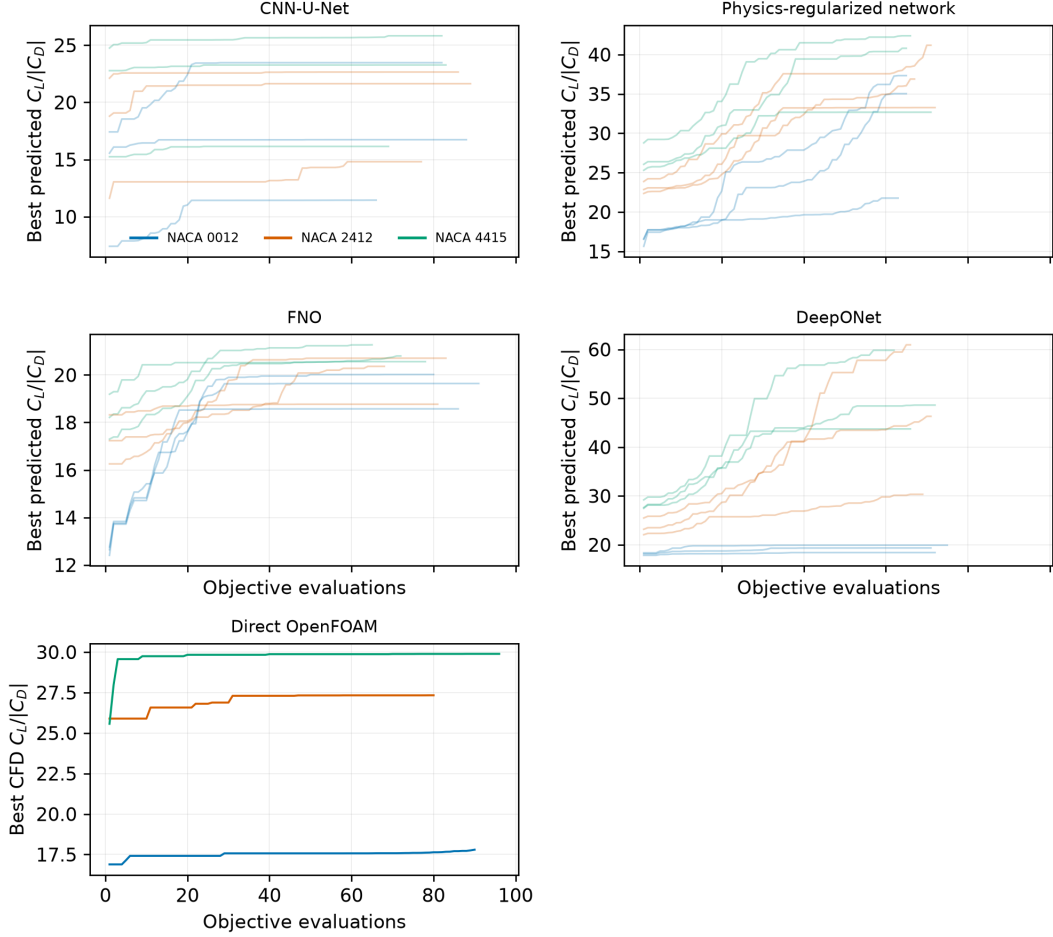


Fig. 6. Best-so-far L/D for all three starts and three training seeds per surrogate; the direct OpenFOAM panel shows its three starts. Every objective evaluation is retained in the released CSV traces.

The water operating conditions also clarify the cavitation result. At $Re = 5 \times 10^5$ and the stated 1 m chord, $U_\infty = 0.5$ m/s and the dynamic pressure is only about 125 Pa. With nominal ambient pressure near 101 kPa, none of the optimized designs approaches vapor pressure; cavitation is therefore reported diagnostically rather than included in the objective. The lower-pressure sweep tests whether the networks preserve the leading-edge low-pressure pattern under an inception criterion; it should not be interpreted as a prediction of a developed vapor cavity. A future cavitation-constrained design study would choose operating pressure and velocity jointly so that inception is active within the search space.

Finally, three surrogate winners approach the lower thickness bound. This is a common place for learned-objective error to matter because the optimizer is querying continuous shapes that were not simulated during training. The fresh mesh-and-solve endpoint check answers the immediate question—whether the selected shape still performs in the reference solver—without claiming that the learned surface is uniformly accurate around that

point. Local CFD samples around each winner, structural thickness constraints, or a trust-region update would be natural additions when the objective is a deployable design rather than the controlled benchmark considered here.

The comparison remains limited to two-dimensional steady RANS and the NACA four-digit space. Repeated seeds and selected-design grid checks quantify two important uncertainties here; external experimental force validation and multiphase cavitation data remain beyond the present benchmark.

VII. Conclusion

We followed four neural surrogates from RANS-field prediction to a design decision using a fixed family-level test set and three seeds. DeepONet gives the lowest mean pressure, C_p , lift, and drag errors; the much smaller physics-regularized network is close and fastest. Complete-field inference is 384–867 times faster than the measured OpenFOAM runs.

The HSO experiment changes the ranking from a simple accuracy table. The physics-regularized network and DeepONet produce the strongest mean medium-grid

endpoints, but DeepONet varies more across seeds. The matched direct-CFD search reaches 29.89, showing that a fast surrogate can change not only search cost but also which basin a budgeted local optimizer reaches. The grid study also shows that converged iterations do not imply converged drag; medium-to-fine L/D shifts remain 16–20%. A scientific surrogate should therefore be judged on final-test prediction, seed robustness, the consequence of acting on its output, and the numerical uncertainty of the verifying simulator.

Artifact Availability

Code, configurations, split definitions, per-case metrics, optimization traces, and exact execution commands are public at <https://github.com/aaravvd/hydrofoil-neural-surrogates>. The CFD dataset and trained checkpoints are archived at <https://doi.org/10.5281/zenodo.21845241>. The repository records the software environment and checksums needed to connect the source release to the archived artifacts.

Acknowledgment and AI-Tool Disclosure

OpenAI Codex was used throughout manuscript preparation as a general-purpose writing and code-assistance tool, including language editing, plotting-script development, and reproducibility checks [22]. The author reviewed the manuscript, reran the reported analyses, and accepts full responsibility for the scientific claims and submitted text.

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